

CSA12 - Computer Architecture

Question 4: 8085/8086 Instruction Simulator

Problem Statement

Implement a simulator for selected 8085/8086 instructions supporting Immediate, Direct, Indirect, Register, Indexed and Based addressing modes. Execute user-defined assembly instructions, update registers and flags, and display memory changes after execution.

Objective

To understand how 8085/8086 processors execute instructions, use different addressing modes, and update registers, flags, and memory.

Theory

An instruction simulator mimics the execution of assembly language instructions. The processor fetches, decodes, and executes instructions while updating registers, memory, and status flags. Common Registers: Accumulator (A) B, C, D, E, H, L Registers Program Counter (PC) Stack Pointer (SP) Status Flags: Carry (CY) Zero (Z) Sign (S) Parity (P) Auxiliary Carry (AC)

Addressing Modes

Addressing Mode	Description	Example
Immediate	Data is part of instruction	MVI A,25H
Direct	Address specified directly	LDA 2500H
Indirect	Address stored in register pair	MOV A,M
Register	Operand in register	MOV A,B
Indexed (8086)	Address + Index Register	MOV AX,[SI+100]
Based (8086)	Base Register + Offset	MOV AX,[BX+10]

Algorithm

1. Read assembly instruction.
2. Decode opcode and addressing mode.
3. Execute instruction.
4. Update registers and flags.
5. Display memory changes.
6. Repeat until END instruction.

Complete C Program

```
#include<stdio.h>
#include<string.h>

int main()
{
    char instruction[20];
    int A=0,B=10;
```

```

printf("Enter Instruction (MOV/ADD/SUB): ");
scanf("%s",instruction);

if(strcmp(instruction,"MOV")==0)
{
    A=B;
    printf("MOV Executed\n");
}
else if(strcmp(instruction,"ADD")==0)
{
    A=A+B;
    printf("ADD Executed\n");
}
else if(strcmp(instruction,"SUB")==0)
{
    A=A-B;
    printf("SUB Executed\n");
}
else
{
    printf("Invalid Instruction");
}

printf("\nAccumulator = %d",A);

return 0;
}

```

Sample Input

ADD

Sample Output

ADD Executed

Accumulator = 10

Time & Space Complexity

Time Complexity: $O(1)$

Space Complexity: $O(1)$

Applications

- Microprocessor education
- Embedded systems
- Assembly language simulation
- Processor debugging

Advantages

- Demonstrates processor instruction execution.
- Easy understanding of addressing modes.
- Useful for laboratory experiments.

Viva Questions

What is an opcode? - The operation part of an instruction.

What is an addressing mode? - Method of accessing operands.

What is the accumulator? - Main register for arithmetic operations.

Name two flags in 8085. - Carry and Zero flags.

What is the difference between 8085 and 8086? - 8085 is an 8-bit processor, while 8086 is a 16-bit processor.